Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
	(c)		Substitution into $PE = \frac{kQq}{r}$ e.g. $\frac{9\times10^{9}\times2.4\times10^{-9}\times1.6\times10^{-19}}{0.6}$ (1) $\frac{9\times10^{9}\times2.4\times10^{-9}\times-1.6\times10^{-19}}{0.6} + \frac{9\times10^{9}\times2.4\times10^{-9}\times-1.6\times10^{-19}}{0.6} + \frac{9\times10^{9}\times2.4\times10^{-9}\times1.6\times10^{-19}}{0.6}$ seen or equivalent or $\frac{9\times10^{9}\times2.4\times10^{-9}\times-1.6\times10^{-19}}{0.6}$ seen with statement that the other 2 potentials cancel (1)	1	1		2	1	
	(d)		Valid method e.g. considering potential energy/potential (1) New PE = $2 \times -5.76 \times 10^{-18} + 2.88 \times 10^{-18} = -8.64 \times 10^{-18}$ [J] **OR** new potential = 54 [V] **OR** PE must be lower because -ve values are the same but the positive is smaller (doesn't need a calculation) **OR** new potential is higher than original because +ve potentials are the same but negative is smaller (36V / 37.5 [V] to 54 [V]) (1) Can travel to lower PE **OR** will gain KE **OR** can travel to higher potential because electron is negative (1) Correct conclusion based on correct physics (yes is sufficient if the explanation is clear) (1) **Alternative based on forces:** Attractive forces (or forces due to +ve charges) are the same/symmetrical for both points OR oscillating with SHM if attractive forces only OR would just reach R with attractive forces only OR force at R is 7.2×10^{-18} [N] ($\rightarrow$) (1) Repulsive force / force due to negative charge helps OR repulsive force at end is 2.4×10^{-18} [N] OR force at P is 19.2×10^{-18} [N] ($\leftarrow$) (1) Overall of these is to the left (1) Conclusion – yes (1)			4	4	2	

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